



Project Title: Financial Institution Network Designed and Implementation

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Project Title: Financial Institution Network Designed and Implementation

1.Objectives

This proposal outlines the design and implementation of a secure, scalable, and efficient network infrastructure for a financial institution. The project will ensure and security across all departments, including secure site-to-site connectivity, wireless access, and robust access control mechanisms. The network setup encompasses various elements including serial router configurations, password security measures and DHCP implementation to streamline IP address management.

2. Requirements

- **Design Tool:** Cisco Packet Tracer.
- **ISPs:** Connect to at least two ISPs for redundancy.
- **Wireless Access:** Deploy wireless networks for each department.
- **VoIP:** Configure IP phones for internal communication.
- **VLANs:** Create department-specific VLANs; assign VLAN 120 for voice traffic across the network.
- **Subnetting:** Allocate IP addresses based on network needs.
- **Inter-VLAN Routing:** Enable communication between VLANs using multilayer switches.
- **Core Switches:** Multilayer switches to handle routing and switching.
- **DHCP:** Use a centralized DHCP server for dynamic IP allocation.
- **Basic Settings:** Configure hostnames, passwords, SSH, encrypted credentials, and banners.
- **OSPF:** Deploy OSPF for dynamic routing between routers and switches.
- **Security:** Apply switchport security for critical segments like the server-side department.

3. Description of System:

Main Branch: Router 1 serves as the gateway for this area. It is responsible for managing network traffic and providing internet access to administrative devices.

Entry Area: Connected to the Main Branch, this area is the focal point for guest interactions. Devices in the reception area connect to the Main Branch router for network services.

Financial Department: Router 2 is connected to this department, applying STICK VLAN segmentation to isolate its network traffic. It ensures efficient communication within the marketing department while maintaining security. IT Department Also connected to Router 7, the IT department has its STICK VLAN to separate its network operations. This setup allows the IT team to manage their resources independently.

Server Room: Router 3 connects this critical area to the central router. It ensures that servers hosting essential hotel services are accessible to all relevant departments while maintaining security measures.

4. Network Design

Network Topology

The network topology comprises the following components:

1. **Routers:** Facilitate interconnection between VLANs and provide gateway access to the internet.
2. **Switches:** Serve to connect devices within each VLAN segment.
3. **PCs:** Representing guest devices including laptops, smartphones.
4. **Web Server:** Hosts essential services such as guest portal, room service requests, etc.
5. **DHCP Server:** Facilitates automatic IP address assignment to network devices.

5. Implementation Steps

Device Configuration

- Assign **hostnames**, **console passwords**, and **banner messages**.
- Enable **SSH** for secure management access.
- Disable **IP domain lookup** to reduce command mis entries.

VLANs

- Create VLANs for data and voice for each department.

- Assign **access ports** to respective VLANs.
- Configure **trunk ports** to support inter-VLAN traffic.

Subnetting and IP Addressing

- Subnets for each department, ensuring sufficient host addresses.
- Assign static IPs to server-room devices and management interfaces.

Inter-VLAN Routing

- Configure inter-VLAN routing on multilayer switches.
- Use **IP helper-address** to relay DHCP requests.

Wireless Network

- Configure SSIDs for each department.
- Secure wireless traffic with WPA2-Enterprise.

VoIP

- Set up the Cisco 2811 router for telephony services.
- Configure IP phones to communicate within their VLANs.

DHCP Server

- Configure a centralized DHCP server.
- Exclude static IP ranges for servers and management devices.

Switchport Security

- Restrict the number of MAC addresses on critical ports.
- Enable violation mode to protect against unauthorized access.

6. Testing and Verification

Connectivity

- Verify VLAN communication using **ping** and **tracert**.
- Test OSPF routes using **show ip route** and **debug ip ospf events**.

Wireless Access

- Confirm wireless SSID visibility and connectivity.
- Test WPA2 authentication for users.

VoIP

- Test inter-department voice calls using IP phones.

DHCP

- Verify dynamic IP assignment using **ipconfig** on client devices.

Security

- Simulate unauthorized access attempts and verify port security actions.

7.Configure Router-01

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.1 255.255.255.0
Router(config-if)#clock rate 56000
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.10.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.12.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

```

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Router-2

```
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#interfac
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed stat
Router(config-if)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#ip add 192.168.14.1 255.255.255.0
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.16.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.15.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

VLAN Configuration

IOS Command Line Interface

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name it
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br

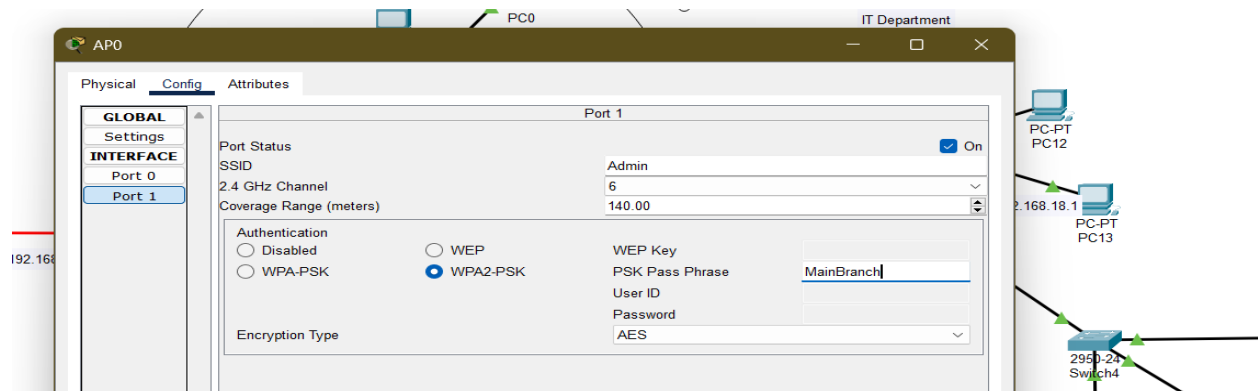
VLAN Name                Status    Ports
-----
1    default                active    Fa0/7, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24
10   marketing              active
20   it                     active
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/4-6
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br
```

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Main Branch



DHCP Implementation

Router 1

```
Router#en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool lan1
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#network 192.168.10.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#
```

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Router 2

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool lan2
Router(dhcp-config)#default-router 192.168.16.1
Router(dhcp-config)#network 192.168.16.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp pool lan3
Router(dhcp-config)#default-router 192.168.15.1
Router(dhcp-config)#network 192.168.15.2 255.255.255.0
Router(dhcp-config)#exit
```

8.Results

- All devices are successfully configured and communicate securely.
- VLAN segmentation and inter-VLAN routing ensure traffic isolation and connectivity.
- OSPF provides dynamic and efficient routing between network devices.
- Wireless networks and VoIP services operate seamlessly.
- Redundant ISP connections ensure failover functionality.

9. Conclusion

The designed network infrastructure achieves the financial institution's objectives by ensuring:

1. **Scalability:** Future-proof design with adequate room for growth.
2. **Security:** VLANs, SSH, and port security prevent unauthorized access.
3. **Reliability:** Redundancy mechanisms maintain uninterrupted service.

The implementation of these features equips the financial institution with a robust, secure, and efficient network that supports its critical operations.

10. Recommendations

- Regularly update firmware and software on all network devices.
 - Periodically test redundancy and failover systems.
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11. Appendices

- Packet Tracer project file (attached).
- Configuration scripts (included in supplementary materials).